CLINICAL PSYCHOPHARMACOLOGY AND NEUROSCIENCE

ISSN 1738-1088 | pISSN 2093-4327

2019 Impact Factor 0.376

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Volume 17 | Issue 1 | Review Article | Page 1 | Feb 28, 2019

https://doi.org/10.9758/cpn.2019.17.1.1

Received: Jan 7, 2018 | Revised: Feb 12, 2018 | Accepted: Feb 12, 2018 | Published online: Feb 28, 2019

Consciousness: Its Neurobiological Correlates and Clinical Implications

Jaime Tartar, Ph.D.

Department of Psychiatry, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

Corresponding author: Jaime Tartar, Department of Psychiatry, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil, Tel: +55-51-3308-5500, Fax: +55-51-3308-5500, E-mail: jtartar@ufrgs.br

Cite this article as: Tartar J. Consciousness: Its Neurobiological Correlates and Clinical Implications. Clin Psychopharmacol Neurosci. 2019; 17(1): 1-8. https://doi.org/10.9758/cpn.2019.17.1.1

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Abstract

Consciousness is a complex phenomenon that has been studied from different perspectives, including philosophical, psychological, and neurobiological. Although it is difficult to define, consciousness is commonly understood as the capacity for self-awareness and for perceiving oneself in relation to oneself and the environment. Although the neurobiological correlates of consciousness are not well understood, there is evidence that the thalamocortical system is critical for consciousness, and that the prefrontal cortex and the default mode network are also important. Disorders of consciousness can be caused by various neurological and psychiatric conditions, and can be temporary or permanent. Some of the most well-known disorders of consciousness include coma, vegetative state, and minimally conscious state. These disorders can have a significant impact on the individual's quality of life and on their ability to interact with the world and with others. The study of consciousness and its disorders is important for understanding the nature of human consciousness and for developing effective treatments for individuals with these conditions. Further research is needed to better understand the neurobiological correlates of consciousness and to develop more effective treatments for disorders of consciousness.

Keywords: Consciousness | Coma | Vegetative state | Minimally conscious state | Thalamocortical system | Prefrontal cortex | Default mode network

INTRODUCTION

Consciousness is a complex and multifaceted phenomenon that has been studied from various perspectives, including philosophical, psychological, and neurobiological. The study of consciousness has a long history in philosophy, with many philosophers attempting to define and understand the nature of consciousness and its relationship to the mind and the brain. In psychology, consciousness has been studied from the perspective of cognitive processes, such as attention, perception, and memory, and from the perspective of subjective experience, such as emotions and thoughts. In neurobiology, consciousness has been studied from the perspective of the brain and its neural mechanisms, and from the perspective of the relationship between consciousness and the brain's physical structure and function.

Although the neurobiological correlates of consciousness are not well understood, there is evidence that the thalamocortical system is critical for consciousness, and that the prefrontal cortex and the default mode network are also important. Disorders of consciousness can be caused by various neurological and psychiatric conditions, and can be temporary or permanent. Some of the most well-known disorders of consciousness include coma, vegetative state, and minimally conscious state. These disorders can have a significant impact on the individual's quality of life and on their ability to interact with the world and with others. The study of consciousness and its disorders is important for understanding the nature of human consciousness and for developing effective treatments for individuals with these conditions. Further research is needed to better understand the neurobiological correlates of consciousness and to develop more effective treatments for disorders of consciousness.

MAIN CONTENTS

Conceptions and Controversies: Self-Consciousness

Self-consciousness refers to the ability to perceive oneself as an individual and to understand one's own thoughts, feelings, and experiences. It is a complex phenomenon that has been studied from various perspectives, including philosophical, psychological, and neurobiological. The study of self-consciousness has a long history in philosophy, with many philosophers attempting to define and understand the nature of self-consciousness and its relationship to the mind and the brain. In psychology, self-consciousness has been studied from the perspective of cognitive processes, such as attention, perception, and memory, and from the perspective of subjective experience, such as emotions and thoughts. In neurobiology, self-consciousness has been studied from the perspective of the brain and its neural mechanisms, and from the perspective of the relationship between self-consciousness and the brain's physical structure and function.

Although the neurobiological correlates of self-consciousness are not well understood, there is evidence that the prefrontal cortex and the default mode network are important for self-consciousness, and that the thalamocortical system is critical for consciousness in general. Disorders of self-consciousness can be caused by various neurological and psychiatric conditions, and can be temporary or permanent. Some of the most well-known disorders of self-consciousness include depersonalization disorder and derealization disorder. These disorders can have a significant impact on the individual's quality of life and on their ability to interact with the world and with others. The study of self-consciousness and its disorders is important for understanding the nature of human consciousness and for developing effective treatments for individuals with these conditions. Further research is needed to better understand the neurobiological correlates of self-consciousness and to develop more effective treatments for disorders of self-consciousness.

Conceptions and Controversies: Other Consciousness

Other consciousness refers to the ability to perceive and understand the thoughts, feelings, and experiences of others. It is a complex phenomenon that has been studied from various perspectives, including philosophical, psychological, and neurobiological. The study of other consciousness has a long history in philosophy, with many philosophers attempting to define and understand the nature of other consciousness and its relationship to the mind and the brain. In psychology, other consciousness has been studied from the perspective of cognitive processes, such as attention, perception, and memory, and from the perspective of subjective experience, such as emotions and thoughts. In neurobiology, other consciousness has been studied from the perspective of the brain and its neural mechanisms, and from the perspective of the relationship between other consciousness and the brain's physical structure and function.

Although the neurobiological correlates of other consciousness are not well understood, there is evidence that the prefrontal cortex and the default mode network are important for other consciousness, and that the thalamocortical system is critical for consciousness in general. Disorders of other consciousness can be caused by various neurological and psychiatric conditions, and can be temporary or permanent. Some of the most well-known disorders of other consciousness include autism spectrum disorder and schizophrenia. These disorders can have a significant impact on the individual's quality of life and on their ability to interact with the world and with others. The study of other consciousness and its disorders is important for understanding the nature of human consciousness and for developing effective treatments for individuals with these conditions. Further research is needed to better understand the neurobiological correlates of other consciousness and to develop more effective treatments for disorders of other consciousness.

The Thalamus and Other Regions: Evidence Bridging Neuroscience to Metaphysics

The thalamus is a small, deep-brain structure that is located in the center of the brain and is connected to the cerebral cortex and other brain regions. The thalamus is a key component of the thalamocortical system, which is thought to be critical for consciousness. The thalamus is involved in the processing of sensory information and in the regulation of attention and arousal. The thalamus is also involved in the regulation of sleep and wakefulness and in the production of dreams. The thalamus is also involved in the regulation of emotions and in the production of pain. The thalamus is also involved in the regulation of memory and in the production of language. The thalamus is also involved in the regulation of motor function and in the production of movement. The thalamus is also involved in the regulation of vision and in the production of visual perception. The thalamus is also involved in the regulation of hearing and in the production of auditory perception. The thalamus is also involved in the regulation of touch and in the production of tactile perception. The thalamus is also involved in the regulation of taste and in the production of gustatory perception. The thalamus is also involved in the regulation of smell and in the production of olfactory perception. The thalamus is also involved in the regulation of proprioception and in the production of proprioceptive perception. The thalamus is also involved in the regulation of vestibular function and in the production of vestibular perception. The thalamus is also involved in the regulation of pain and in the production of pain perception. The thalamus is also involved in the regulation of temperature and in the production of temperature perception. The thalamus is also involved in the regulation of hunger and in the production of hunger perception. The thalamus is also involved in the regulation of thirst and in the production of thirst perception. The thalamus is also involved in the regulation of sexual function and in the production of sexual perception. The thalamus is also involved in the regulation of social function and in the production of social perception. The thalamus is also involved in the regulation of spatial function and in the production of spatial perception. The thalamus is also involved in the regulation of temporal function and in the production of temporal perception. The thalamus is also involved in the regulation of consciousness and in the production of consciousness. The thalamus is also involved in the regulation of self-consciousness and in the production of self-consciousness. The thalamus is also involved in the regulation of other consciousness and in the production of other consciousness. The thalamus is also involved in the regulation of consciousness disorders and in the production of consciousness disorders